

DNA molecule synthesis

Part 1: Iterative Synthesis of ACTCTGAT

Step 1.1: Coupling Reaction (First Nucleotide Addition: A)

Option 1.1.1: TdT-mediated incorporation of 3'-ONH₂-dATP in phosphate buffer.

- Component A: Immobilized initiator oligonucleotide on beads (at 1 μ M final concentration)
- Component B: ZsTdT-R335L-K337G (at 2 mg/mL final concentration)
- Component C: 3'-ONH₂-dATP (at 0.50 mM final concentration, purified as in Part 2)
- Component D: CoCl₂ (at 0.50 mM final concentration)
- Component E: phosphate buffer (pH 6.8) (at 50 mM final concentration)
- Component F: NaCl (at 100 mM final concentration)

(Operational details: Combine all components, ensuring they are prepared with HPLC-grade water and free from aldehyde/ketone contamination. Incubate the reaction mixture at 30 °C for 5 min with gentle tilting and rotation to keep beads suspended.)

Step 1.2: Post-Coupling Wash

Option 1.2.1: Wash beads with 1X Binding & Washing (B&W) Buffer.

- Component A: Beads from Step 1.1
- Component B: 1X B&W Buffer
 - Recipe: 5 mM Tris-HCl (pH 7.5), 0.5 mM EDTA, 1 M NaCl, 0.1% Tween-20

(Operational details: Use a magnetic stand to separate the beads for 1 minute. Remove the supernatant. Add 1X B&W Buffer, resuspend the beads, separate, and discard the supernatant. Repeat wash cycles 2-3 times.)

Step 1.3: Deblocking Reaction

Option 1.3.1: Cleavage of the 3'-ONH₂ protecting group using sodium nitrite in acetate buffer.

- Component A: Washed beads from Step 1.2
- Component B: Cleavage Reagent
 - Recipe: 700 mM NaNO₂ in 1 M aqueous sodium acetate buffer, pH 5.5

(Operational details: Resuspend the beads in the Cleavage Reagent and incubate at room temperature for 5 minutes with gentle rotation. This step may be repeated to ensure complete cleavage.)

Step 1.4: Post-Deblocking Wash

Option 1.4.1: Wash beads with 1X Binding & Washing (B&W) Buffer to remove deblocking agent.

- Component A: Beads from Step 1.3
- Component B: 1X B&W Buffer
 - Recipe: 5 mM Tris-HCl (pH 7.5), 0.5 mM EDTA, 1 M NaCl, 0.1% Tween-20

(Operational details: Use a magnetic stand to separate the beads for 1 minute. Remove the deblocking buffer. Add 1X B&W Buffer, resuspend the beads, separate, and discard the supernatant. Repeat wash cycles 2-3 times.)

Step 1.5: Subsequent Synthesis Cycles for ACTCTGAT

Option 1.5.1: Repeat steps 1.1 through 1.4 for the remaining 7 nucleotides.

(Operational details: For each cycle, perform the Coupling (Step 1.1), Post-Coupling Wash (Step 1.2), Deblocking (Step 1.3), and Post-Deblocking Wash (Step 1.4). The specific 3'-ONH₂-dNTP used in the coupling step is changed for each cycle according to the target sequence ACTCTGAT.)

- Cycle 2 (C): Use 3'-ONH₂-dCTP.
- Cycle 3 (T): Use 3'-ONH₂-dTTP.
- Cycle 4 (C): Use 3'-ONH₂-dCTP.
- Cycle 5 (T): Use 3'-ONH₂-dTTP.
- Cycle 6 (G): Use 3'-ONH₂-dGTP.
- Cycle 7 (A): Use 3'-ONH₂-dATP.
- Cycle 8 (T): Use 3'-ONH₂-dTTP.

Wash the beads prior to initiating synthesis. Synthesis proceeds in the 5' to 3' direction. Adjust the number of parallel synthesis to satisfy the need of following steps.